Running the original experiment 3 RTL file, the compilation report shows 22 registers. I added 3 registers for the following signals: write_enable_a[0], write_enable_a[1], and write_enable_b[1], which means I should get 25 registers. The compilation report shows 24. I think this is because read_address is only ever incremented by 2, so bit 0 is always 0 and Quartus most likely removed the register read_address[0]. The report supports this: it lists read_address[1] to read_address[8] but no bit 0.

The critical path runs from RAM1's output through the signed comparator and the mux into RAM1's data input. Fmax is 124.72 MHz. It is the longest path because it chains the RAM output delay, the comparison and the mux in one clock cycle. The address counter paths are shorter.

I first drew the circuit on paper. I put W[2m] on the top port of the first RAM, W[2m+1] on its bottom port, X[2m] on the top port of the second RAM, and X[2m+1] on its bottom port. read_address counts up by 2 and is used only for the top ports of both RAMs. write_address starts at 1, counts up by 2, and is used only for the bottom ports of both RAMs. This gives me both 2m and 2m+1. I adapted the read/write states from experiment 2, switching between reading and writing every clock cycle, so that I could use all 4 ports. I put all the combinational logic in an always_comb block, which compares the two values and assigns them to Y and Z as the documentation specifies